LECTURE :36

Sustainable Education

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Community-based ESD practices and policy

Okayama Commitment 2014:Japan Case Study

- **Japan , Okayama Commitments** :
- • Environmental conservation
- • Disaster risk reduction (DRR)
- • Income generation, entrepreneurship, community development
- • Cultural diversity and dialogue, intergenerational exchange
- • Literacy
- • Empowerment

The Six action principles are:

- **Responding to emergent local community issues** that impact on achieving sustainable development
- **Engaging with different key players** to respond effectively to the issues identified
- **Enabling all those who seek solutions** to sustainability issues to contribute actively to finding them
- **Embedding the principles** and practices of **ESD** in daily life
- **Sustaining changes** and achievements beyond short-term project timelines
- **Transforming ourselves**, our practices and policies towards a more sustainable future

The four policy support mechanisms, :

- Providing resources
- Networking/partnership
- Capacity-building
- Content development

How education is related to other SDG targets(UNESCO, 2016a)

Goal 1	Education is critical to lifting people out of poverty.
Goal 2	Education plays a key role in helping people move towards more sustainable farming methods, and in understanding nutrition.
Goal 3	Education can make a critical difference to a range of health issues, including early mortality, reproductive health, spread of disease, healthy lifestyles and well-being.
Goal 5	Education for women and girls is particularly important to achieve basic literacy, improve participative skills and abilities, and improve life chances.
Goal 6	Education and training increase skills and the capacity to use natural resources more sustainably and can promote hygiene.
Goal 7	Educational programmes, particularly non-formal and informal, can promote better energy conservation and uptake of renewable energy sources.
Goal 8	There is a direct link among such areas as economic vitality, entrepreneurship, job market skills and levels of education.
Goal 9	Education is necessary to develop the skills required to build more resilient infrastructure and more sustainable industrialization.

Goal 10	Where equally accessible, education makes a proven difference to social and economic inequality.
Goal 11	Education can give people the skills to participate in shaping and maintaining more sustainable cities, and to achieve resilience in disaster situations.
Goal 12	Education can make a critical difference to production patterns (e.g. with regard to the circular economy) and to consumer understanding of more sustainably produced goods and prevention of waste.
Goal 13	Education is key to mass understanding of the impact of climate change and to adaptation and mitigation, particularly at the local level.
Goal 14	Education is important in developing awareness of the marine environment and building proactive consensus regarding wise and sustainable use.
Goal 15	Education and training increase skills and capacity to underpin sustainable livelihoods and to conserve natural resources and biodiversity, particularly in threatened environments.
Goal 16	Social learning is vital to facilitate and ensure participative, inclusive and just societies, as well as social coherence.
Goal 17	Lifelong learning builds capacity to understand and promote sustainable development policies and practices.

Source: ICSU and ISSC (2015)

- **The Frame-Model for Sustainability Competencies**
- Regarding the goals or objectives recommended for ESD, two major sources were identified.
- 1. There is a need to change attitudes among people so that they are able to assess and address their concerns about sustainable development.
- 2. There are goal recommendations which were mostly developed by researchers from the educational field, and which often achieve a higher degree of differentiation; analysis of existing ESD (learning) goals has revealed a significant deficit in terms of the operationalization of the ESD output.
- 3.Without operationalization, and the resulting non-existent measuring instruments, the needs for ESD and effects of ESD-related interventions (for example lessons, seminars, projects) cannot be determined empirically.

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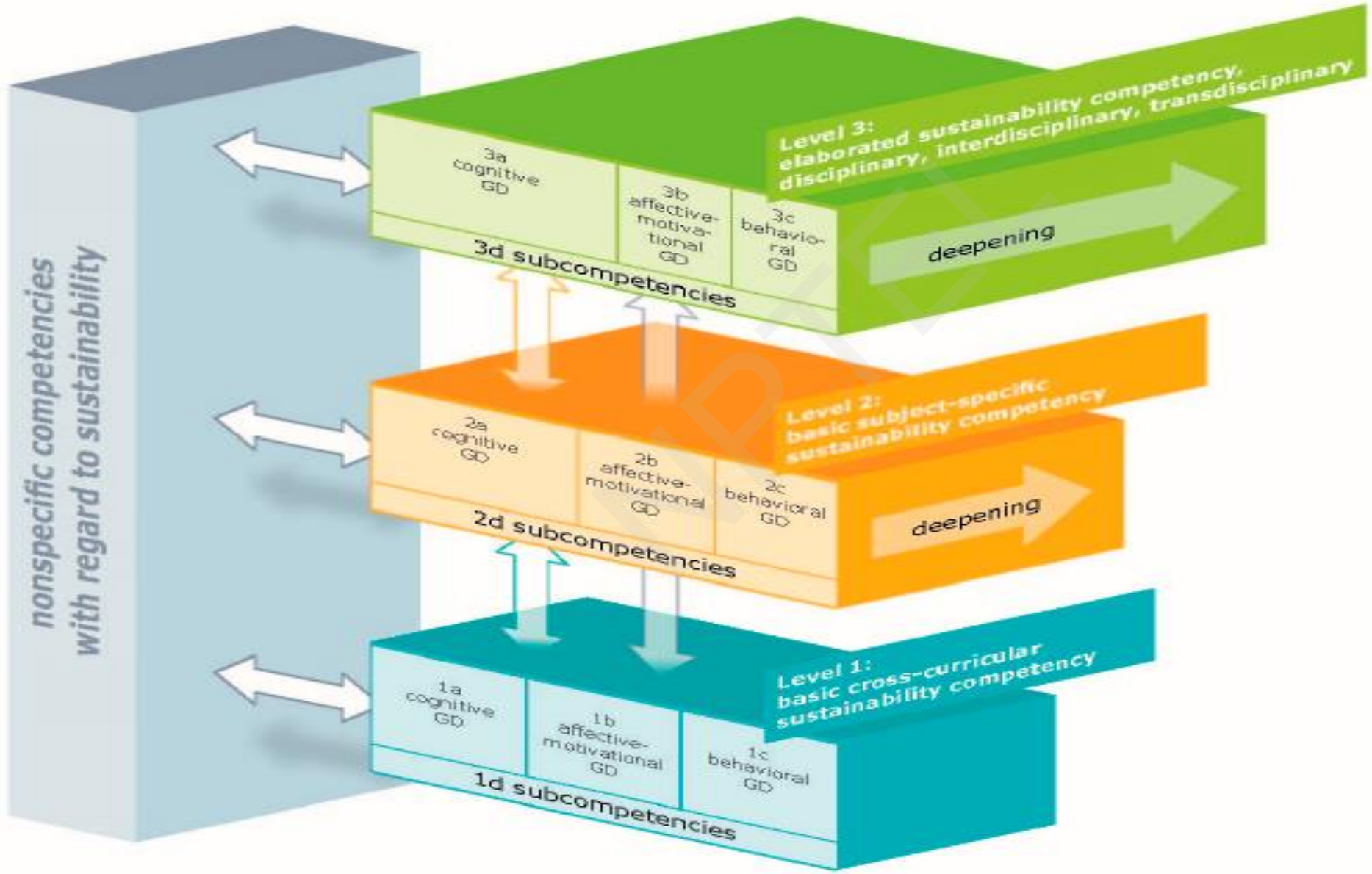
How can this shortcoming be addressed?



- **Frame-model** for the structuring of relevant sustainability competencies with cognitive, affective-motivational, and behavioral goal dimensions (GD), and sub competencies on each level . On the left, the interacting nonspecific competencies (with regard to sustainability)(Source: Sustainability 2019, 11, 1717, 4 of 20).
- In the sense of hot cognitions ,the affective-motivational facets of sustainability competencies (Figure 1, field 1b, 2b, 3b) include all affect-, need-, and motivation-related competency features.
- These include, among other things, values (such as personal acceptance of the intergenerational idea of justice or the personally favored lifestyle), attributions of responsibility or attitudes (for example in questions of consumption or mobility (on Level 1 and 2) and mature epistemological beliefs about the relativity and situational character of empirical knowledge in important areas of sustainability on Level 3).
- The underlying affective-motivational traits become progressively more conscious on the way from Level 1 to Level 3. An affective goal commitment or a positively assessed sequence of actions is the core of any motivation, without which an action will not be performed.

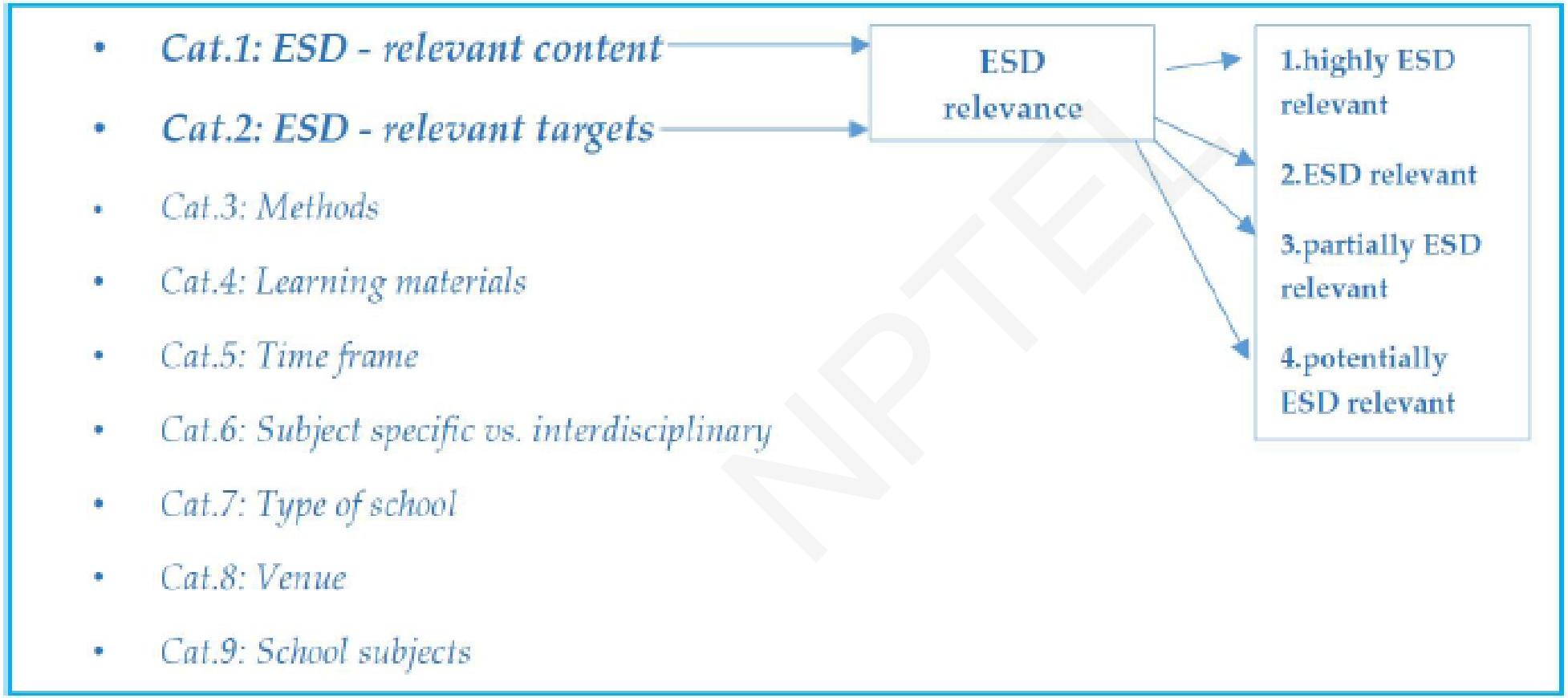
- **German Research Study:**
- Research Question 1: What is the present state of (lower secondary) student sustainability competencies in Germany (Baden-Wuerttemberg) after the implementation of ESD as a new guiding principle?
- Research Question 2: What kinds of subscales can be developed and used to capture sustainability competencies?
- The measurement instrument presented here provides a valuable starting point for further test extension, e.g., for assessing learning progression, to assess the development of competencies in multilevel analysis or longitudinal studies.
- Results revealed some first insights into the interactions of different socio-demographical aspects with the different sustainability competency dimensions.
- In future studies it will be interesting to explore the role of the interactions of the different facets of sustainability competencies in more depth and compare them to other findings of relevant research in the field of for example psychology for sustainability to further explore the interconnections.

Table 1. Samples for each dimension of the questionnaire.

	Questionnaire Dimension	Sample	Item Type
1.	Socio-demographic	*Age, gender, chosen subject specification*	Mixed
2.	Basic cross-curricular and subject-specific sustainability related knowledge	*If someone wants to live sustainably, she/he/* should...* *Please mark only one response option (the most appropriate).* ☐... *always donate money for aid and conservation projects in poorer countries.* ☐... *if possible, eat local farming products, invest money profitably and fight for worldwide peace.* ☐... *eat as vegan as much as possible to protect local animal species, to create a fairer life for animals and humans.* ☐... *respect nature, advocate for justice, and make sure everyone has enough to live on.*	Single-choice items with four response options (one right answer, three distractors) [1]
3.	Affective-motivational beliefs towards sustainability	*When I hear of cars that consume a lot of fuel, I get angry.*	Ordinal scale (4-point Likert, from *Strongly disagree* to *Strongly agree*)
4.	Self-reported sustainability-related behavior intentions	*When I buy chocolate with my pocket money, I buy organic or fair-trade chocolate.*	Ordinal scale (4-point Likert, from *Strongly disagree* to *Strongly agree*)
5.	Intentional subject-specific sustainability knowledge applied *class 5–6 (age group 9–14)* *class 7–8 (age group 11–16)*	*recycling, smart phone production/use, decline in insect numbers* *sustainable products labels/consumption, organic agriculture, renewable energy*	Single-choice items with different response formats (true/false –fast response items)
6.	Dilemma situations	*Imagine the following situation: You are the boss of a big company and you can make all decisions for this company. Your company has earned a lot of money this year. You decide: (Please mark only one response option.)* ☐ *I install a new solar system and an electric power charging station for my employees' e-cars.* ☐ *I build a new sports area so that my staff will feel well and stay healthy.* ☐ *I hire a consultant who helps me with future decisions to make the production more environmentally friendly and improve the working conditions for the employees, and I implement this program.* ☐ *I pay myself a big salary so I have money to go on holidays with my family.*	Single-choice items with four response options on an ordinal ranking [2]

[1] Correct answer is marked; [2] The answer with the highest score covers three sustainability dimensions and is marked.

An ESD Indicator for Teacher Training & the National Monitoring for ESD: Implementation in Germany



Categories of the coding procedure for analyzing the TT course descriptions with nine categories validated and tested by external ESD experts.(Source: Sustainability 2018, 10, 2508 7 of 17)

Thematic indicator framework : Education 2030 Framework for Action
Source: UNESCO

Goal: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

1	2	3	4	5	6	7	8	9	10	11
Concept	No.	Indicator	Equity	Sex	Location	Wealth	Available	Coverage	Proposed global indicators*	Comments
		Targets 4.1- 4.7								
4.1 By 2030, ensure that all girls and boys complete free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes										
Learning	1.	Percentage of children/ young people (i) at the end of primary and (ii) at end of lower secondary achieving at least a minimum proficiency level in (a) reading and (b) mathematics	Yes	X	X	X	1-3 years		Yes	Efforts to generate global measures will focus on common concepts of minimum proficiencies drawing links between national, regional and international student assessment studies.
	2.	Administration of a nationally representative learning assessment (i) during primary (ii) at the end of primary and (iii) at the end of lower secondary education	No				1-3 years			Standards need to be developed for all assessments, national, regional and international to improve their quality. The UIS Observatory of Learning Outcomes will track national, regional and international assessments.
Completion	3.	Gross intake ratio to the last grade (primary, lower secondary)	Yes	X			Yes	c150		
	4.	Completion rate (primary, lower secondary, upper secondary)	Yes	X	X	X	Yes	c100		This indicator is currently available but work is required to finalize a common methodology and increase the number of surveys available to calculate it. The rate is calculated for young people aged 3-5 years above the official ending age for a given level of education.

- **Teacher Education for Sustainable Development in Germany :**
- **Learning objectives for teachers to promote ESD**
- 1. Know about sustainable development, the different SDGs and the related topics and challenges
- 2. Understand the discourse on and the practice of ESD in its local, national and global context
- 3. Develop their own integrative view of the issues and challenges of sustainable development by taking into account the social, ecological, economic and cultural dimensions from the perspective of the principles and values of sustainable development including that of intergenerational and global justice
- 4. Take disciplinary, interdisciplinary and transdisciplinary perspectives on issues of global change and their local manifestations

- Reflect on the concept of sustainable development, the challenges in achieving the SGDs, the importance of their own field of expertise for achieving the SDGs and their own role in this process
- Reflect on the relationship of formal, non-formal and informal learning for sustainable development, and apply this knowledge in their own professional work
- Understand how cultural diversity, gender equality, social justice, environmental protection and personal development are integral elements of ESD and how to make them a part of educational processes
- Practice an action-oriented transformative pedagogy that engages learners in participative, systemic, creative and innovative thinking and acting processes in the context of local communities and learners' daily lives
- Act as a change agent in a process of organizational learning that advances their school towards sustainable development

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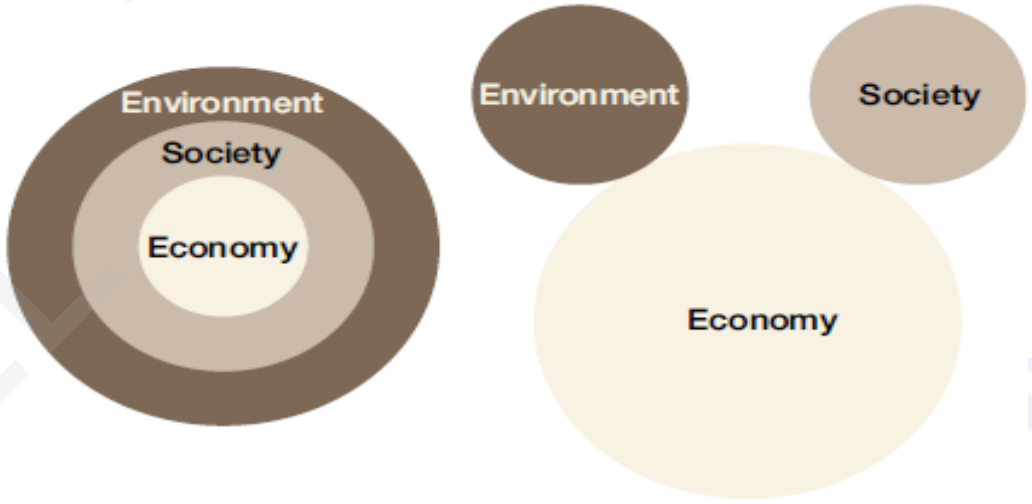
- Identify local learning opportunities related to sustainable development and build cooperative relationships
- Evaluate and assess the learners' development of cross-cutting sustainability competencies and specific sustainability-related learning outcomes
- Possible modules of a teacher education curriculum with ESD as a key element.
- Basic concepts of sustainable development from a local, national and international perspective
- ESD concepts from a local, national and international perspective
- Disciplinary, interdisciplinary and transdisciplinary views of key examples of sustainability challenges
- Project-oriented work on specific problems of local, national and global importance in cooperation with educational institutions and other (local) partners
- Research-based analysis of ESD processes in different learning settings (such as schools, colleges or non-formal educational institutions)
- Practical experiences with ESD approaches and their critical reflection

UNESCO ESD Sourcebook (UNESCO 2012), whole-school approaches



VISUALIZING SUSTAINABILITY

- The formal curriculum contains knowledge, skills, perspectives and values related to sustainability.
- Learning includes real-life issues to enhance pupils' motivation and learning.
- The school has a sustainability ethos, which can be seen in the treatment of others, school property, and the environment.
- School management practices reflect sustainability.
- School policies reflect environmental, social, and economic sustainability.
- Interactions between the school and community are fostered.
- Special events and extra-curricular activities apply and enhance classroom learning about sustainability.
- Pupils engage in decision-making affecting school life.

Monitoring Progress: The SDG India Index

- Goal-wise ranking of states/ UTs and overall ranking based on performance on all goals
- Promotes competition among the states/ UTs
- Enable states/ UTs to learn from peers
- Supports states/ UTs in identifying priority areas
- Highlights data gaps

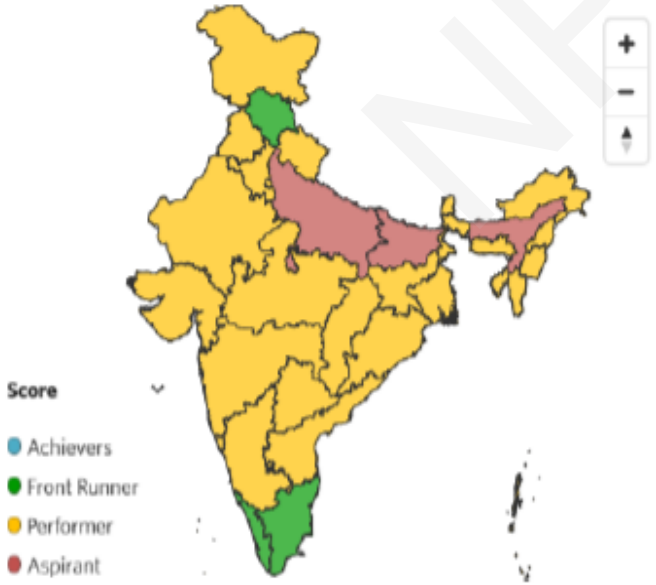
SUSTAINABLE DEVELOPMENT GOALS — 17 GOALS TO TRANSFORM OUR WORLD | NITI Aayog

Ranking | State/UT Performance | Methodology | Metadata

Select Goal

SDG India Index

State Scores



Performance category

	Category	States
1	Achiever	
2	Front Runner	Himachal Pradesh, Kerala, Tamil Nadu, Chandigarh, Puducherry
3	Performer	Andhra Pradesh, Arunachal Pradesh, Chhattisgarh, Goa, Gujarat, Haryana, Jammu and Kashmir, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Punjab, Rajasthan, Sikkim, Telangana,

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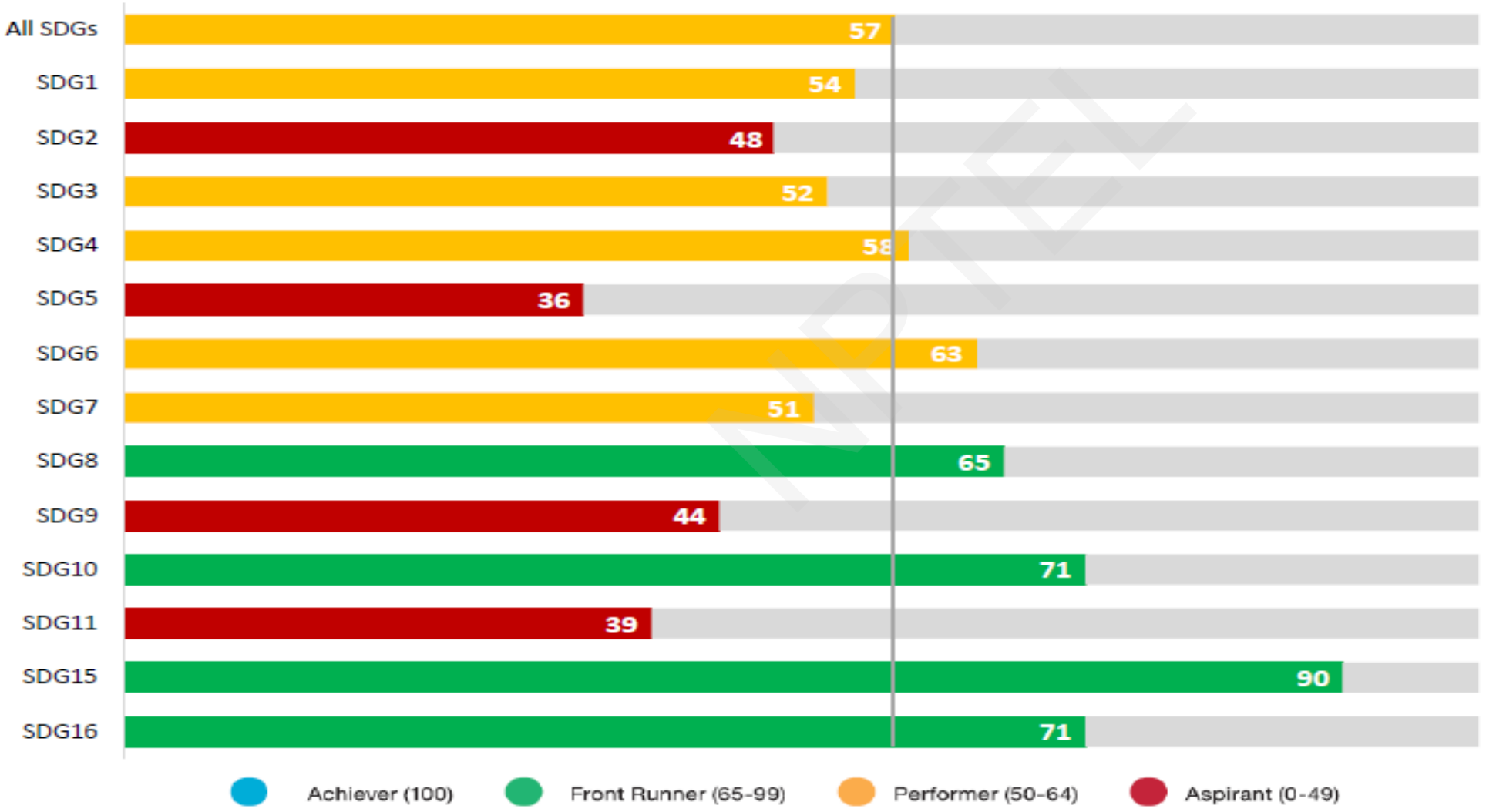
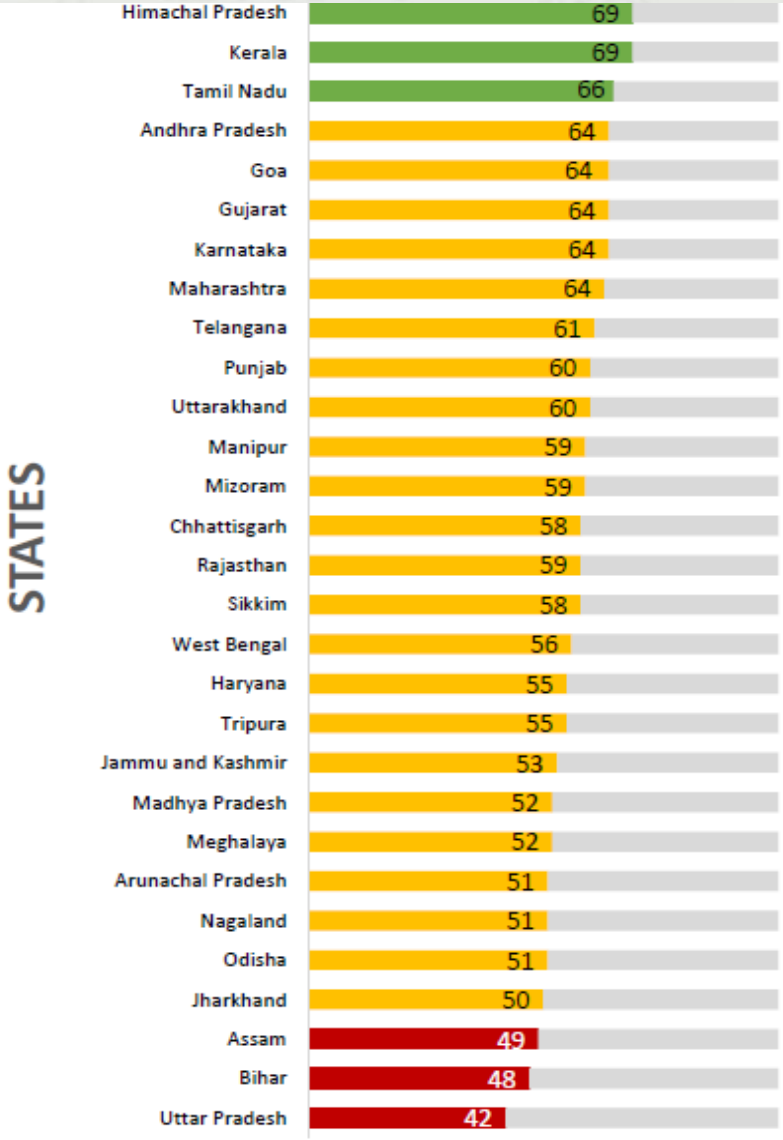
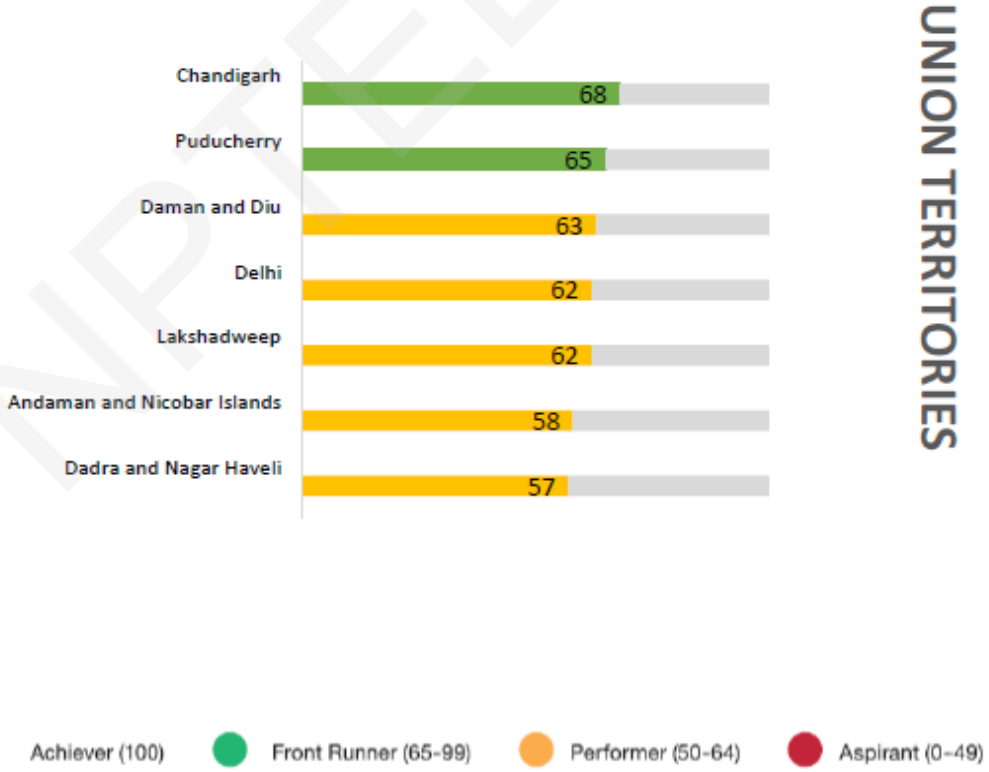
INDIA'S INDEX SCORE: OVERALL AND SDG-WISE



OVERALL SCORE





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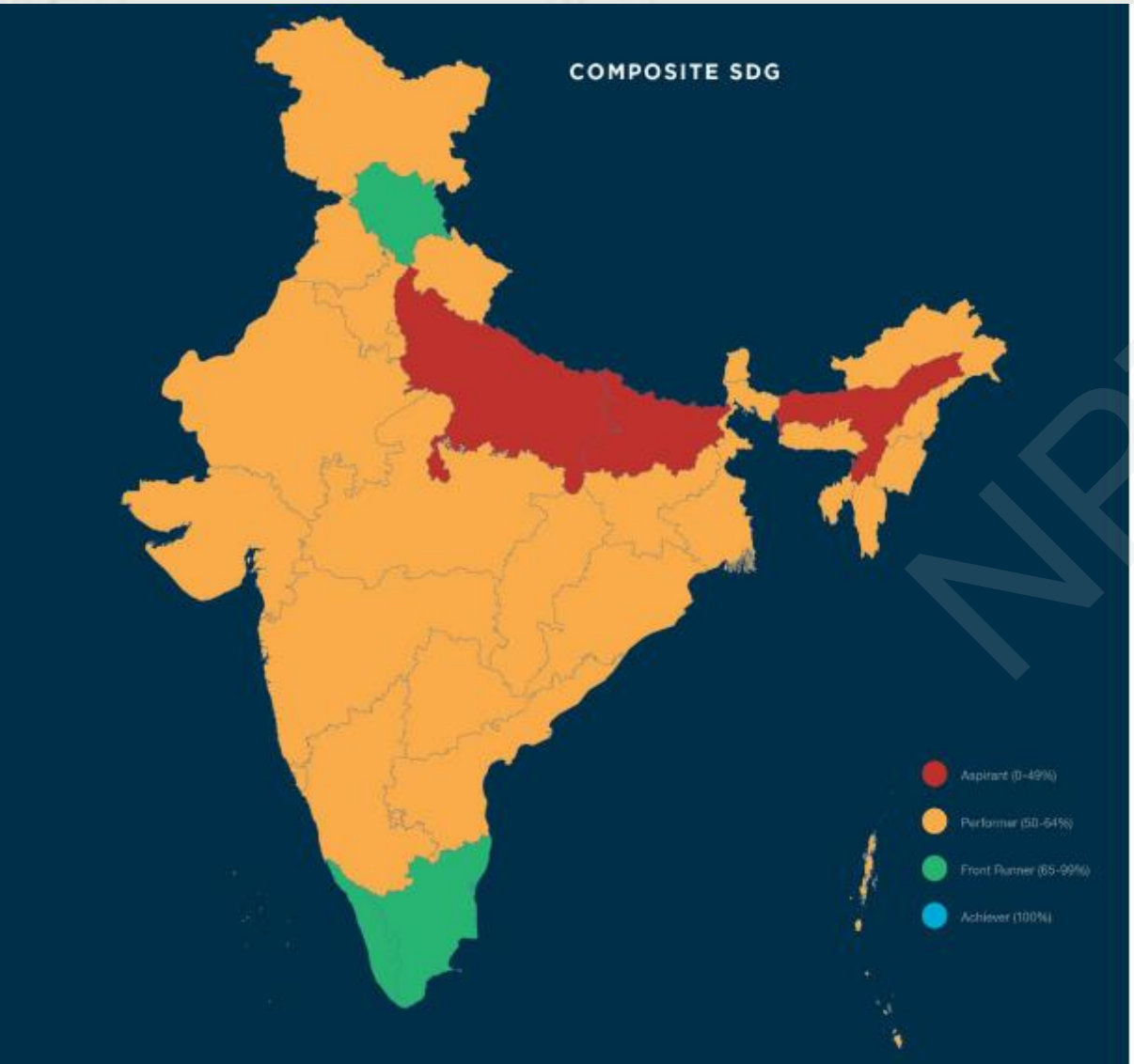

Front Runners

Chandigarh
Himachal Pradesh
Kerala
Puducherry
Tamil Nadu

Performers

A & N Islands
Andhra Pradesh
Arunachal Pradesh
Chhattisgarh
Dadra & Nagar Haveli
Daman & Diu
Goa
Gujarat
Haryana
Jammu & Kashmir
Jharkhand
Karnataka
Lakshadweep
Madhya Pradesh
Maharashtra
Manipur
Meghalaya
Mizoram
Nagaland
Odisha
Punjab
Rajasthan
Sikkim
Telangana
Tripura
Uttarakhand
West Bengal

Aspirants

Assam
Bihar
Uttar Pradesh

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Meeting Challenges…

- Generating data on all SDG Indicators from local to national level
- Convergence and efficiency of financing SDGs across various actors and institutions
- Capacity development of multiple stakeholders on a range of gaps
- Identifying regional focus areas & developing action strategies

Emerging Focus Areas

SDG Financing

- Mobilizing **public finance:** Gap analysis followed by interventions in tax capacity, enhancing spending efficiency and rationalization of spending, assessing fiscal sustainability and space

Engaging the Private Sector

- Mainstreaming SDGs in core business beyond the CSR domain
- Investment in **research and innovation** in new technology and low-cost solutions in areas like waste management, resource efficiency etc.

Strengthening the Data System

- Operationalising the **State and District Indicator Frameworks**
- Functional systems for ensuring data integrity

Leveraging Science, Technology, Innovation

- Using STI to **Leave No One Behind** & deliver quality public services for everyone
- STI platforms to promote **entrepreneurship** & business development

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UNESCO Futures of Education Report:
Reimagining our futures together

https://en.unesco.org/news/united-nations-alert-education-should-be-clear-priority

https://www.youtube.com/watch?v=j8aXSTN71MY

https://en.unesco.org/news/sdg4youth-network-positioning-students-and-youth-front-and-centre-sdg4

https://en.unesco.org/news/global-media-and-information-literacy-week-media-and-information-literacy-public-good

- **Perspectives of Sustainable Development**
- 1. To achieve sustainable development and a higher quality of life for all people, countries should reduce and eliminate unsustainable patterns of production and consumption and promote appropriate demographic policies.
- 2. Women play a vital role in environmental management and development. Their full participation is therefore essential to achieving sustainable development.
- 3. Warfare is inherently destructive to sustainable development. Peace, development and environmental protection are interdependent and indivisible.
- 4.A systems thinking approach , rather than an approach that looks at problems in isolation should be used. Sustainability issues are linked and part of a “whole.”
- 5.Understanding local issues in a global context and recognizing that solutions to local problems can have global consequences.

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- 6.Realizing that individual consumer decisions affect and give rise to resource extraction and manufacturing in distant places.
- 7.Considering differing views before reaching a decision or judgement.
- 8.Recognizing that economic values, religious values, and societal values compete for importance as people with different interests and backgrounds interact.
- 9.Seeing all humans as having universal attributes.
- 10. Knowing that technology and science alone cannot solve all of our problems.
- 11.Emphasizing the role of public participation in community and governmental decision-making. People whose lives will be affected by decisions must be involved in the process leading to the decisions.
- 12.Calling for greater transparency and accountability in governmental decision-making.

- 12.Employing the precautionary principle – taking action to avoid the possibility of serious or irreversible environmental or social harm even when scientific knowledge is incomplete or inconclusive.
- Therefore, educators, leaders, and citizens recognize that sustainable development is an evolving concept and that the list of sustainability perspectives can therefore grow and change.
- Governments and civil society as well as individuals must hold the responsibility for a more sustainable future. All must contribute in their own way.
- **Global Action Programme (GAP) on ESD states that:**
- ESD allows every human being to acquire the knowledge , skills, values and attitudes that empower them to contribute to sustainable development and take informed decisions and responsible actions for environmental integrity, economic viability, and a just society for present and future generations. ESD promotes skills like critical thinking, understanding complex systems, imagining future scenarios, and making decisions in a participatory and collaborative way (UNESCO, 2014b: 33).

Development of sustainability competencies:

The GAP highlights learning outcomes that stimulate learning and promote core competencies, such as 'critical and systemic thinking, collaborative decision-making, and taking responsibility for present and future generations' (UNESCO, 2014b: 12).

Key competencies for sustainable development (Wiek et al., 2016):

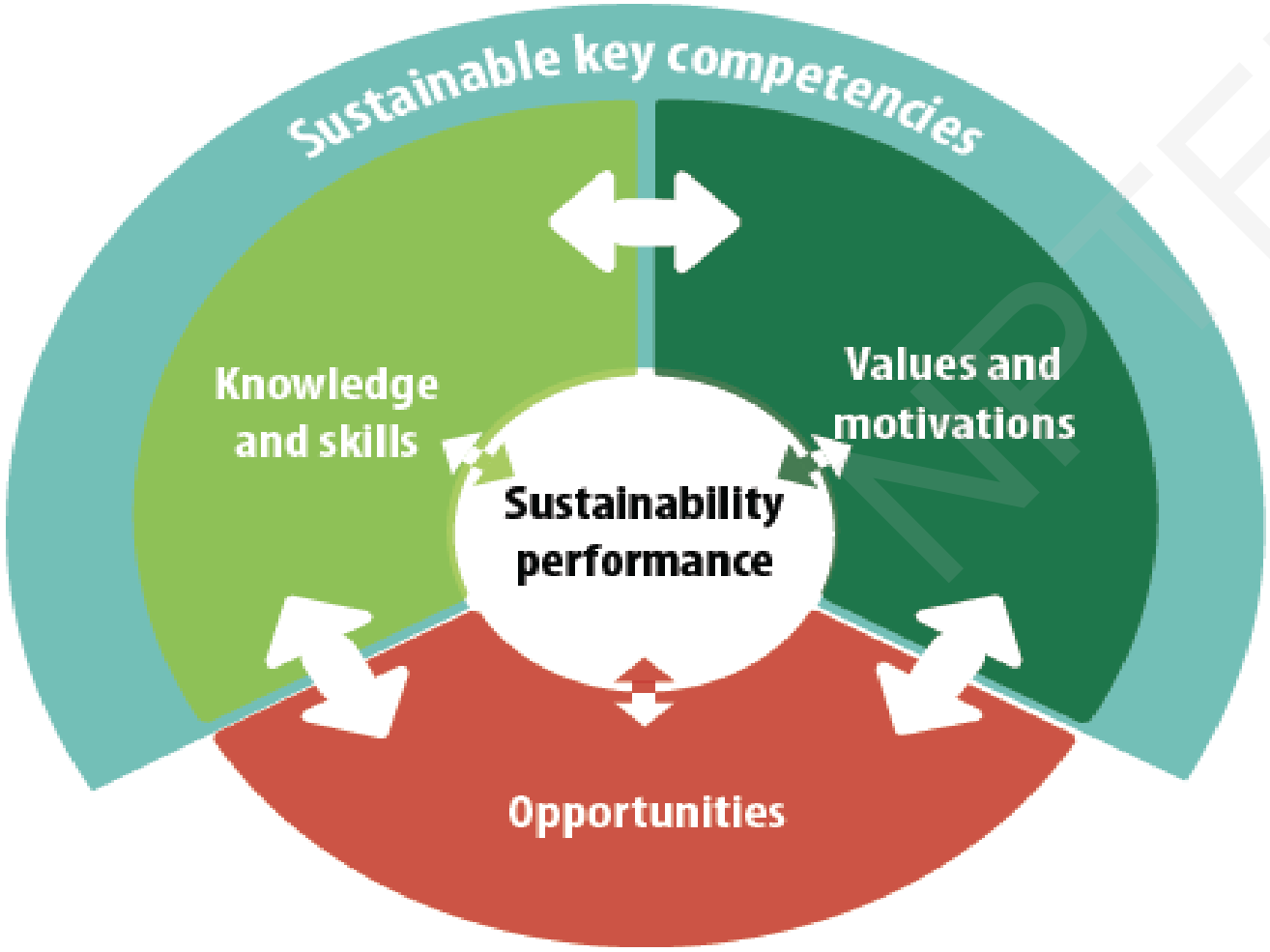
- **Anticipatory competency**: the ability to understand and evaluate multiple futures – possible, probable and desirable – and to create one's own visions for the future, to apply the precautionary principle, to assess the consequences of actions, and to deal with risks and changes;
- **Normative competency**: the ability to understand and reflect on the norms and values that underlie one's actions and to negotiate sustainability values, principles, goals and targets, in a context of conflicts of interests and trade-offs, uncertain knowledge and contradictions;

- • **Strategic competency**: the ability to collectively develop and implement innovative actions that further sustainability at the local level and

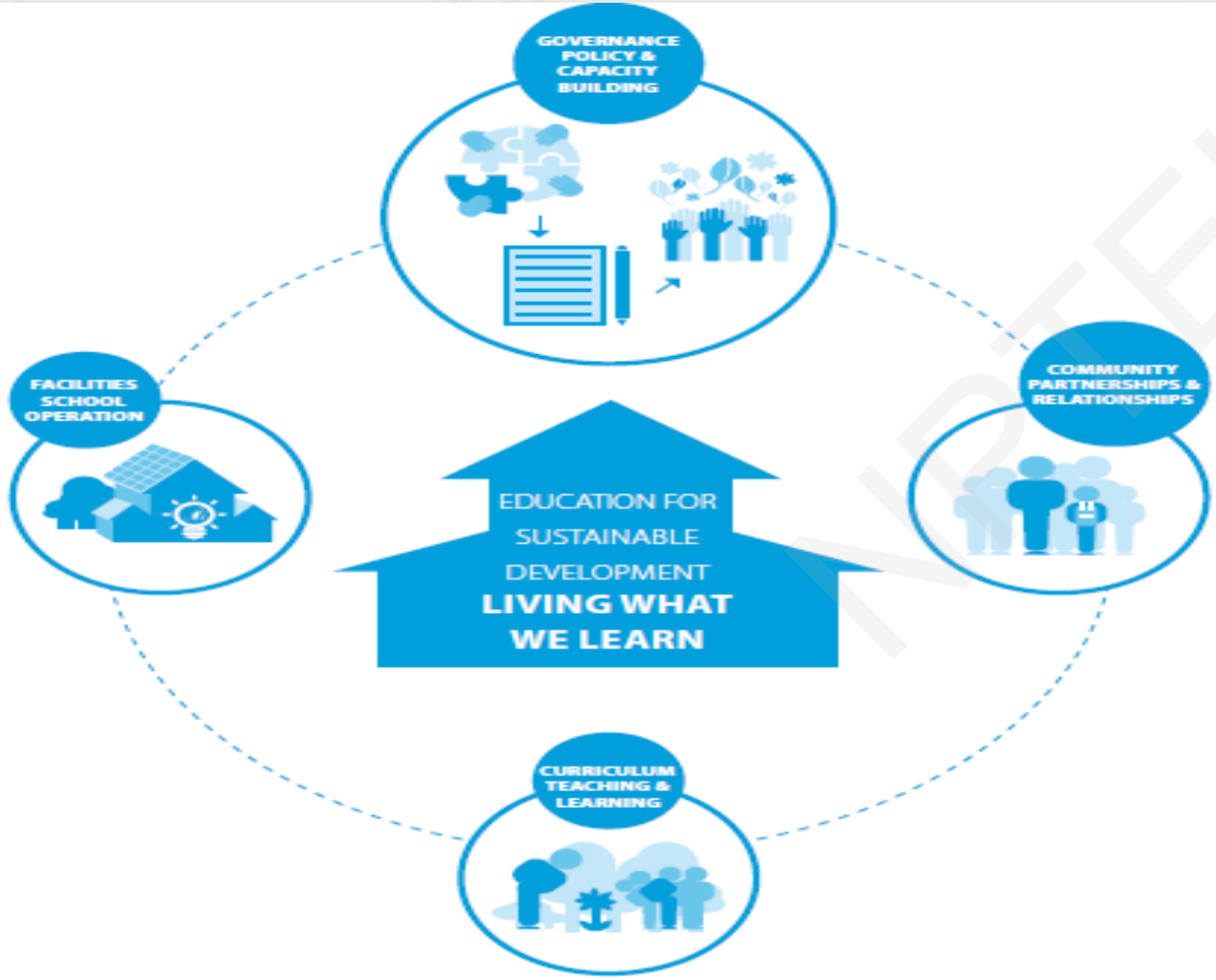
  further a field;
- • **Collaboration competency**: the ability to learn from others; understand and respect the needs, perspectives and actions of others (empathy);understand, relate to and be sensitive to others (empathic leadership),deal with conflicts in a group; and facilitate collaborative and participatory problem-solving;
- • **Critical thinking competency**: the ability to question norms, practices and opinions; reflect on own one's values, perceptions and actions; and take a position in the sustainability discourse;
- • **Self-awareness competency**: the ability to reflect on one's own role in the local community and (global) society, continually evaluate and further motivate one's actions, and deal with one's feelings and desires;
- • **Integrated problem-solving competency**: the overarching ability to apply different problem-solving frameworks to complex sustainability problems and develop viable, inclusive and equitable solution that promote sustainable development – integrating the above-mentioned competencies.

Key competencies and performance of sustainability citizens

Source: (A. Leicht, J. Heiss and W. J. Byun (eds) © UNESCO 2018)



Implications of ESD for the practice of education and pedagogy



Source: UNESCO (2014a: 89).

Whole-institution approach:

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Box 1: Key elements for whole-institution approaches

1. An institution-wide process is organized in a manner that enables all stakeholders – leadership, teachers, learners, administration – to jointly develop a vision and plan to implement ESD in the whole institution.
2. Technical and, where possible and appropriate, financial support is provided to the institution to support its reorientation. This can include the provision of relevant good practice examples, training for leadership and administration, the development of guidelines, as well as associated research.
3. Existing relevant inter-institutional networks are mobilized and enhanced in order to facilitate mutual support such as peer-to-peer learning on a whole-institution approach, and to increase the visibility of the approach to promote it as a model for adaptation.

Source: UNESCO (2014b).

- While all elements of the whole-institution approach are important, interactive, integrative and critical forms of learning are at the core of delivering ESD in the classroom and other learning settings, making this approach an action-oriented transformative pedagogy.
- **Action-oriented transformative pedagogy**
- ESD is about developing sustainability competencies and, thus, empowering and motivating learners to become active and critical sustainability citizens able to participate in shaping a sustainable future. The pedagogical approaches needed to achieve this end should be learner-centred action oriented and transformative.
- **Key pedagogical approaches in ESD:**
- **1. A learner-centred approach**
- Learner-centred pedagogy sees students as autonomous learners and emphasizes the active development of knowledge rather than its mere transfer and/or passive learning experiences. The learners' prior knowledge as well as their experiences in the social context are the starting points for stimulating learning processes in which the learners construct their own knowledge base.

Learner-centred approaches require learners to reflection their own knowledge and learning processes in order to manage and monitor them. Educators should stimulate and support those reflections . Learner-centred approaches change the role of an educator from that of an expert who transfers structured knowledge to that of a facilitator of learning processes (Barth, 2015).

2.Action-oriented learning

In action-oriented learning, learners engage in action and reflect on their experiences in relation to the intended learning process and personal development. The experience might come from a project (e.g. in-service learning), an internship, facilitation of a workshop, implementation of a campaign and so on. Action-learning draws on Kolb's learning cycle of experimental learning, which has the following stages:

(i) having a concrete experience, (ii) observation and reflection, (iii) formation of abstract concepts for generalization and (iv) application in new situations (Kolb, 1984).

contd…

Action learning increases knowledge acquisition, competency development and values clarification by linking rather abstract concepts to personal experience and the learners' life. The role of the educator is to create a learning environment that prompts learners' experiences and reflexive thought processes.

- **3.Transformative learning**
- Transformative learning can be defined primarily by its aims and principles , not by a concrete teaching or learning strategy. It aims to empower learners to question and change their ways of seeing and thinking about the world , in order to further develop their understanding of it (Mezirow , 2000; Slavich and Zimbardo, 2012). The educator acts as a facilitator who empowers and challenges learners to change their worldviews. The related concept of transgressive learning (Lotz-Sisitka et al., 2015) goes one step further – it states that learning in ESD has to overcome the status quo and prepare the learner for disruptive thinking and the co-creation of new knowledge.